SUMMARY

1. Engineered Features

Eight engineered features were created to provide the models with additional information beyond the original Adult Income dataset variables. These were designed to capture nonlinear relationships, interactions, and meaningful combinations of the original variables.

Engineered Feature	Description
age_group	Groups individuals into meaningful career/age categories
hours_group	Groups weekly working hours into work-intensity categories
capital_net	Represents the net effect of capital gain and capital loss
has_capital_gain	Binary indicator showing whether capital gain is present
has_capital_loss	Binary indicator showing whether capital loss is present
education_age_ratio	Relates education level to age
work_hours_age	Combines working hours with age
is_married	Binary indicator representing married status

2. Cross-Validation Comparison

Three models were evaluated using 5-fold cross-validation on F1 score and ROC AUC. HistGradientBoosting achieved the strongest results on both metrics.

Model	F1 Mean	F1 Std	ROC AUC Mean	ROC AUC Std
Logistic Regression	0.6693	0.0035	0.9127	0.0036
Random Forest	0.6677	0.0018	0.9024	0.0023
HistGradientBoosting	**0.7063**	0.0069	**0.9259**	0.0039

3. Statistical Comparison

The two best models by cross-validated F1 — HistGradientBoosting and Logistic Regression — were compared using a paired Wilcoxon signed-rank test on their five corresponding cross-validation F1 scores.

- HistGradientBoosting mean F1: **0.7063**
- Logistic Regression mean F1: **0.6693**
- Difference: **0.0370 F1 points**
- Wilcoxon statistic: **0.0**
- p-value: **0.0625**

Since the p-value exceeds 0.05, the difference is not statistically significant at the 5% level. However, HistGradientBoosting shows a practically noticeable improvement of roughly 0.037 F1 points; the small number of CV folds limits the test's statistical power.

4. Feature Importance

Permutation importance for HistGradientBoosting ranks is_married as the most influential engineered feature, followed by education_age_ratio and work_hours_age.

Engineered Feature	HGB Permutation Importance
is_married	**0.017906**
education_age_ratio	0.005916
work_hours_age	0.003956
hours_group	0.002306
age_group	0.001866
capital_net	0.000662
has_capital_gain	0.000000
has_capital_loss	**0.000000**

For Logistic Regression, the largest positive coefficients among the engineered numeric features were capital_net (+2.069) and is_married (+0.766). These results are broadly sensible marital status, education, age, working hours, and capital-related information are all expected to relate to income. The zero permutation importance of the two binary capital indicators suggests they add little beyond the original capital variables for HistGradientBoosting.

5. Feature Selection & Recommendation

SelectKBest with mutual information was used to reduce the 121 preprocessed features to the top 60 features.

Feature Set	CV F1 Mean	F1 Std	Training Time
All 121 features	**0.7063**	0.0069	5.32 sec
Selected 60 features	0.6986	0.0074	**1.31 sec**

Feature selection cut training time by roughly 75%, but F1 dropped by about 0.0077. Since F1 is the primary evaluation metric, this performance loss outweighs the computational savings.

Recommended Model & Features for Tomorrow

Primary model: HistGradientBoosting

Features: Retain all 121 preprocessed features, including the engineered features.

HistGradientBoosting achieved the highest cross-validated F1 (0.7063) and ROC AUC (0.9259). Although feature selection substantially reduced training time, it reduced F1 performance, so the full feature representation will be used for hyperparameter tuning.

Secondary model: Logistic Regression is retained as a comparison baseline (second-highest F1, 0.6693).

Tomorrow's hyperparameter tuning will focus on HistGradientBoosting using all 121 processed features.